CS F364 : Design & Analysis of Algorithms

Project Report

A Faster Algorithm for Betweenness Centrality

Group 26

Group Members

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1 Individual Contributions

- **Kriti Saluja (ID: 2022B5A70698H):** Implemented the Brandes' algorithm based on the reference paper.
- **M Adithya Krishna (ID: 2023A7PS0144H):** Worked on executing the implementation on the provided datasets and verifying correctness of outputs.
- **Mriganka Thakur(ID: 2022B5A71309H):** Assisted in testing and improving the performance of the implementation through optimizations.
- **Vishal Goel (ID: 2023A7PS1108H):** Contributed to writing and structuring the project report.
- **Veer Gandhi (ID: 2023A7PS1234H):** Assisted in documentation and preparation of the final report.

2 System Configuration

All experiments were conducted on a Linux-based system with the following configuration:

- Operating System: Linux (Pop_OS)
- Processor: 13th Gen Intel(R) Core(TM) i5-13420H
- RAM: 16 GB
- CPU Utilization: 99%
- Compiler: `g++` with optimization flags `-O3 -march=native`

3 Output of the Program

3.1 Dataset 1: Email-Enron

Graph Statistics:

- Number of Nodes: 36,692
- Number of Edges: 367,662

Top 20 Nodes by Betweenness Centrality:

```
Node 5038 : 4.36511e+07
Node 140 : 4.06606e+07
Node 566 : 2.4451e+07
Node 588 : 2.38678e+07
Node 1139 : 2.38324e+07
Node 273 : 2.35422e+07
Node 458 : 2.32279e+07
Node 46 : 2.06187e+07
Node 1028 : 1.90793e+07
Node 292 : 1.71775e+07
Node 195 : 1.71215e+07
Node 370 : 1.69549e+07
Node 823 : 1.42478e+07
Node 136 : 1.39262e+07
Node 76 : 1.35066e+07
```

```
Node 286 : 1.27319e+07
Node 734 : 1.25411e+07
Node 543 : 1.254e+07
Node 1768 : 1.25335e+07
Node 353 : 1.16238e+07
```

Execution Time:

- Algorithm Time: 161.18 seconds
- User Time: 169.73 seconds
- Wall-clock Time: 2 min 49.87 sec
- CPU Usage: 99%

Memory Usage:

- Approximate Memory (Code Estimate): 3.82 MB
- Maximum Resident Set Size (RSS): 19,168 KB

3.2 Dataset 2: Wiki-Vote

Graph Statistics:

- Number of Nodes: 7,115
- Number of Edges: 103,689

Top 20 Nodes by Betweenness Centrality:

```
Node 2565 : 1.63467e+06
Node 11 : 877073
Node 457 : 870049
Node 4037 : 719298
Node 1549 : 719283
Node 1166 : 637643
Node 766 : 610759
Node 15 : 509771
Node 1374 : 481392
Node 2237 : 390004
Node 1151 : 359944
Node 2688 : 324125
Node 2328 : 322901
Node 2470 : 298823
Node 737 : 280109
Node 5524 : 261900
Node 3352 : 259205
Node 1186 : 253139
Node 72 : 248580
Node 5079 : 242407
```

Execution Time:

- Algorithm Time: 8.80 seconds
- User Time: 8.79 seconds

- Wall-clock Time: 8.80 seconds
- CPU Usage: 99%

Memory Usage:

- Approximate Memory (Code Estimate): 1.0 MB
- Maximum Resident Set Size (RSS): 7,832 KB

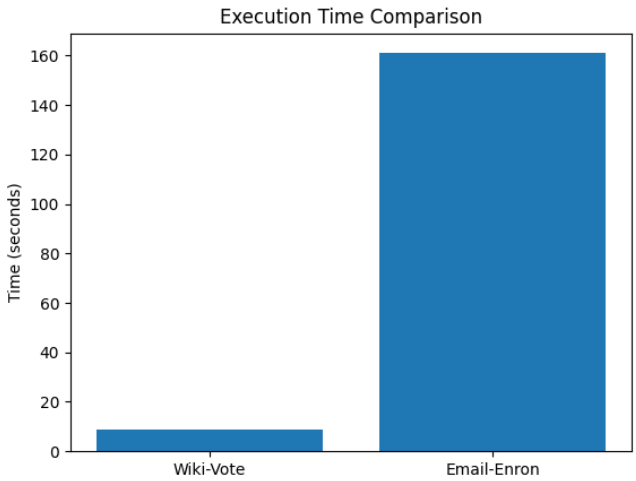

Figure 1: Execution Time Comparison between Wiki-Vote and Email-Enron datasets